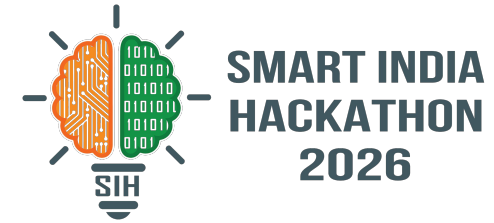


SMART INDIA HACKATHON 2026



- **Problem Statement ID – 26153**
- **Problem Statement Title - AI based Network Attack Forecasting from Network Traffic Data**
- **Theme - Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID -**
- **Team Name - Foresight**



1 Detailed Explanation of the Proposed Solution

Beyond Static IDS

Traditional classifiers evaluate isolated packets/flows ($X \rightarrow Y$), missing low-and-slow campaigns



Reactive & Limited View

Network World Model

Learns the environment's state-transition dynamics: $P(S_{t+1} | S_t)$ mapping how network telemetry evolves over time.



$$P(S_{t+1} | S_t)$$

Model the "physics" of network behavior

Autoregressive K-Step Rollout

Performs forward simulation K time windows ahead to predict infiltration risk timelines.



Visualize future risk before lateral movement or exfiltration

Dual Ingestion Engine

Ingests both NetFlow/IPFIX flow statistics and PCAP packet-level metrics.



+



- Port scan entropy
- TTL variance
- TCP window dynamics
- Retransmissions
- Flow aggregates

2 How It Addresses the Problem



Proactive Lead Time

Shifts defense from post-breach detection to pre-compromise forecasting, providing defenders a **3.5 to 7.0-minute** mitigation window.

3.5 – 7.0 minutes

EARLY WARNING → FASTER RESPONSE



Kill Chain Context

Classifies evolving traffic into explicit MITRE ATT&CK stages.

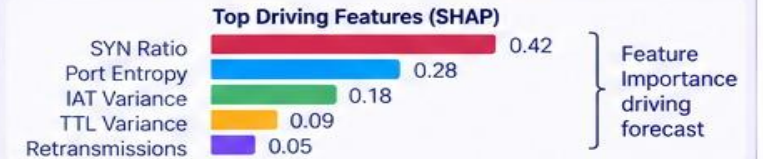


END-TO-END ATTACK LIFECYCLE UNDERSTANDING



Transparent Decision Support

Integrates SHAP attribution to highlight the exact telemetry driving the forecast (e.g., SYN ratio spikes, port entropy).



3 Innovation and Uniqueness



First-of-its-Kind State Transition Engine

Replaces black-box classifiers with a physical transition operator T that predicts future network features.

$$S_{t+1} = T(S_t)$$

- ✓ Causal understanding
- ✓ Predictive capability
- ✓ Not just classification



Multi-Task Objective

Concurrently optimizes state dynamics reconstruction (MSE) alongside tactic classification using Class-Balanced Sparse Focal Loss.

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \cdot \mathcal{L}_{\text{Focal}}$$

- ✓ State reconstruction
- ✓ Tactic classification
- ✓ Handles class imbalance (99:1)



100% Air-Gapped / Sovereign

Operates completely offline with zero cloud API dependencies for classified/CII networks.



1 Technologies to be Used



Languages & Core

- Python 3.10+
- NumPy
- Pandas



Packet & Flow Processing

- Scapy (session reconstruction, port entropy)
- PyShark
- NetFlow parser



Deep Learning Engine

- TensorFlow / Keras
- Conv1D + Bidirectional LSTM + Multi-Head Dense



Explainability (XAI)

- SHAP GradientExplainer
- Tactic head attribution



Demonstration UI & Deployment

- Streamlit (offline dashboard)
- Plotly Express
- Docker & Docker-Compose

2 Implementation Methodology & Architecture Pipeline

1 Telemetry Ingestion



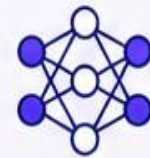
- Raw PCAP (Packet Level)
- Flow CSVs (CSE-CIC-IDS2018)

2 Temporal State Matrix



- Resample into 10s/30s bins
- Stack to $X \in \mathbb{R}^S(T \times F)$ ($T = 15$)

3 Multi-Task World Model



- Conv1D (Micro-burst patterns)
- BiLSTM (Macro kill chain)
- State + Attack + Tactic Heads

4 Forward Simulation (K Steps)



- Autoregressive state rollout
- Risk score timeline projection
- MITRE ATT&CK phase mapping

5 Interpretability (SHAP)



- Driving feature attribution
- Entropy / SYN / IAT weights

6 Air-Gapped Streamlit UI



- Infiltration curve & lead time
- MITRE ribbon & flow tables

1 FEASIBILITY ANALYSIS



Algorithmic Feasibility
benchmark datasets
(CIC-IDS 2018, UNSW-NB15)



Computational Efficiency
1D CNN + BiLSTM
runs on standard CPU/edge
hardware.



Operational Fit
Integrates with
NetFlow/IPFIX and PCAP
formats; leverages
firewalls and network taps.

2 POTENTIAL CHALLENGES & STRATEGIES FOR OVERCOMING

! IDENTIFIED RISK / CHALLENGE	OPERATIONAL IMPACT	MITIGATION STRATEGY
 Extreme Class Imbalance	Benign flows dominate 99% of enterprise traffic.	 Implemented Sparse Focal Loss ($\gamma=2.0$) and inverse-frequency class weights to heavily penalize missed attacks.
 Autoregressive Error Drift	Multi-step rollout (K-steps) can accumulate error.	 Constrained forecast horizon (K=5 to 15 windows); bounded state predictions using batch normalization.
 High Throughput PCAP Load	Packet inspection can create computational bottlenecks.	 Implemented dual-mode ingestion: fast flow-level monitoring with lightweight statistical packet aggregation (port entropy, TTL).
 Concept Drift & Novel Exploits	Attackers alter tools and signatures.	 Dynamic Knowledge Graph mapper (CVE → CWE → CAPEC → ATT&CK) to generalize by tactics, not signatures.

1 POTENTIAL IMPACT ON TARGET AUDIENCE



Security Operations Centers (SOC):

Replaces alert fatigue with prioritized, forward-looking trajectory forecasts.



Critical Information Infrastructure (CII):

Protects defense, energy, banking, and telecom networks where downtime or breach is catastrophic.



Incident Responders:

Delivers granular SHAP attributions explaining why an attack is progressing, speeding up mean-time-to-respond (MTTR).

2 MEASURABLE BENEFITS (OPERATIONAL, ECONOMIC & SECURITY)



Quantifiable Metric Improvements (vs. Static Baseline):



Macro F1-Score:

Reaches **90.3%** vs. 71.0% on static baseline (+**27.1%** relative improvement).



False Positive Reduction:

Drops false alarm rate from **8.4%** to **1.6%** (over **80%** reduction in false alarms).



Defensive Lead Time:

Generates **3.5 to 7.0 minutes** of actionable warning, enabling automated SOAR playbooks to quarantine IPs before data exfiltration.



Economic Advantage:

Prevents **multi-crore** breach remediations and regulatory penalties by stopping intrusions in the reconnaissance and initial access phases.



National Security Alignment:

Strengthens **India's cyber resilience** and **indigenous self-reliance** (*Atmanirbhar Bharat*) using open-source, air-gapped sovereign AI.

1 RESEARCH PAPERS & FOUNDATIONS



- 1 Ha, D., & Schmidhuber, J. (2018). *World Models*. Advances in Neural Information Processing Systems (NeurIPS).



- 2 Sharafaldin, I., et al. (2018). *Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization (CIC-IDS2018)*. Communications in Computer and Information Science.



- 3 Lin, T. Y., et al. (2017). *Focal Loss for Dense Object Detection*. IEEE Transactions on Pattern Analysis and Machine Intelligence (Applied to extreme network imbalances).



- 4 Lundberg, S. M., & Lee, S. I. (2017). *A Unified Approach to Interpreting Model Predictions (SHAP)*. NeurIPS.

2 OPEN CYBERSECURITY KNOWLEDGE BASES & DATASETS

	MITRE ATT&CK Enterprise Matrix	 https://attack.mitre.org/
	CAPEC (Common Attack Pattern Enumeration and Classification)	 https://capec.mitre.org/
	NIST National Vulnerability Database (NVD / CVE)	 https://nvd.nist.gov/
	Stratosphere Laboratory CTU-13 Dataset	 https://stratosphereips.org/datasets-ctu13
	UNSW-NB15 Dataset	 Cyber Range Lab of the Australian Centre for Cyber Security (ACCS)



GitHub Prototype

Explore the project source code and prototype on GitHub.

<https://github.com/gagan-long/SIH2026-AI-Network-Attack-Forecaster->

